# Martian Map: Project Pre-Plan

*A dual-profile route planner for human and rover explorers on Mars*

## 1. Title and Team

- Project name: Mertialoy [ *Dual-Profile Route Planner for Human and Rover Explorers*]
- Challenge: Interplanetary Survival Guide: Martian Map (theme: The Next Frontier)
- Team members: Md. Imran Hasan, Md. Masum Parvej, Maksumul Haque, Afrazur Rahman
- One-line pitch: Draw a route on Mars and get a verdict: whether it is safe, how much energy it costs, and what science lies along the way.

## 2. Team Roles

| Member | Role | Responsibility |
|---|---|---|
| Md. Imran Hasan | Team Lead | architecture, Git workflow and code review, integration testing, deadlines, final pitch. |
| Afrazur Rahman | Technical | map interface, route drawing, hazard overlay, survival card, offline deployment, GitHub repo. |
| Md. Masum Parvej | Data Engine | elevation data (DEM) processing, slope, route segmentation, energy model, explorer profiles. |
| Maksumul Haque | Science and Research | Reference constants and citations, documentation, project page, videos |

## 3. Problem and Challenge Statement

NASA has explored Mars robotically for decades, and the first astronauts to walk there will need the best possible map. Today, the information needed to plan a surface route (terrain, energy cost, hazards and science targets) sits in separate datasets and tools. The challenge asks for a layered, integrated view of a Martian location or route that draws on multiple NASA science missions and helps an explorer plan and carry out a safe walk while doing new science along the way.

**Research question:** How can one shared route engine judge the same route under two different energy models, electrical-only for a rover and suit power plus metabolic effort for a human?

**Gap:** Most route tools are either rover-centred or terrain-centred. Few place human and rover explorers side by side with separate energy accounting and a science layer in a single view.

## 4. Objectives

1. Compute distance, slope and travel time for any route drawn within one chosen Martian region (for example Jezero Crater).
2. Provide two explorer profiles with separate energy ledgers: electrical energy in watt-hours for a rover, and suit electrical energy (Wh) plus metabolic energy (kcal) for a human. The two are never merged into one number.
3. Flag hazards on each route segment: steep slopes, rough or rocky terrain, soft ground, and distance from a safe return point (with exposure limits for humans).
4. Validate the tool by comparing its output with at least one real rover traverse.

5. Present each route as a survival card showing pass or fail, energy used, safe stopping points and the reason for any failure.
6. Deliver a complete, public, documented submission on time .

## 5. Solution Approach and Overview

The solution is a website, not a simulation. The user selects a region, draws a route and chooses an explorer type. The engine then computes the route against real NASA data and returns a judgment.

- **Flow:** select region, draw route, choose profile, compute, check hazards, show survival card.
- **Explorer profile:** a set of parameters (mass, base power draw, climb cost, speed, slope limit and energy type) that describes who or what is travelling. A rover and a human astronaut are two profiles run through the same engine.
- **Design principle:** the engine contains no rover-versus-human branching; all differences come from the selected profile.
- **Output:** a survival card per route, with a safer alternative suggested when a route fails (stretch goal).

## 6. Scope

**Minimum viable product (must ship):**

- One region with cached terrain data
- Route drawing, with slope, distance and time computed
- Dual energy model with the Human and Rover toggle
- Hazard flags and the survival card
- On-screen data citations and full offline operation

**Stretch goals (only after the MVP is complete):**

- Science-interest layer: real rover discoveries and mineral signatures near the route
- Contingency route suggestion when a route fails
- Bangla language toggle

**Out of scope:**

- Covering the whole planet, real-time data, live rover control, and medical-grade human physiology

## 7. Data Sources

All sources are open NASA data. Each must be verified against the official challenge statement released on 28 October 2026.

| Data | Used for | Source |
|---|---|---|
| Elevation and terrain | Slope, distance | NASA Mars Trek |
| High-resolution imagery | Rough terrain, visual context | HiRISE, via the Planetary Data System |
| Mineral maps | Science points | CRISM, via the Planetary Data System |
| Rover mission records | Science points, validation | NASA mission pages and PDS |
| Rover specifications | Rover profile constants | NASA mission fact sheets |

| Data | Used for | Source |
|---|---|---|
| Spacesuit and metabolic data | Human profile constants | NASA human spaceflight reports |

Every constant used in the profiles is recorded with its source and link in a single reference sheet, so judges can check it against our claims.

## 8. Method and Calculations

- **Slope:** angle computed from adjacent elevation points along the route.
- **Distance and time:** segment length divided by explorer speed, with speed adjusted by slope.
- **Energy:** base power multiplied by time, plus climb cost multiplied by elevation gain. Rover: one electrical ledger. Human: suit electrical ledger and metabolic ledger, each checked against its own limit.
- **Hazard rules:** slope above the profile limit, rough-terrain score, return distance beyond the safe energy budget, and exposure limits for humans.
- **Assumptions:** simplified physics, Mars gravity of about 0.38 g, and constant environmental conditions. These are listed openly.
- **Validation:** compare computed results with a real rover traverse, and test usability with a few non-expert users.
- **Limitations:** the model is simplified and is not intended for mission-grade or medical use.

## 9. Impact and Novelty

- **Novelty:** one tool that handles both human and rover explorers with separate, honest energy models, plus a science layer on the same route.
- **Impact:** faster early-stage route screening for planners and researchers, and an accessible educational tool for students and the public.
- **Alignment with judging criteria:** Impact (route safety and reach), Creativity (dual-profile design), Validity (comparison with real traverses), Relevance (NASA data at the core), and Presentation (the survival card demo).